

Feature Evolution in Diffusion Language Models

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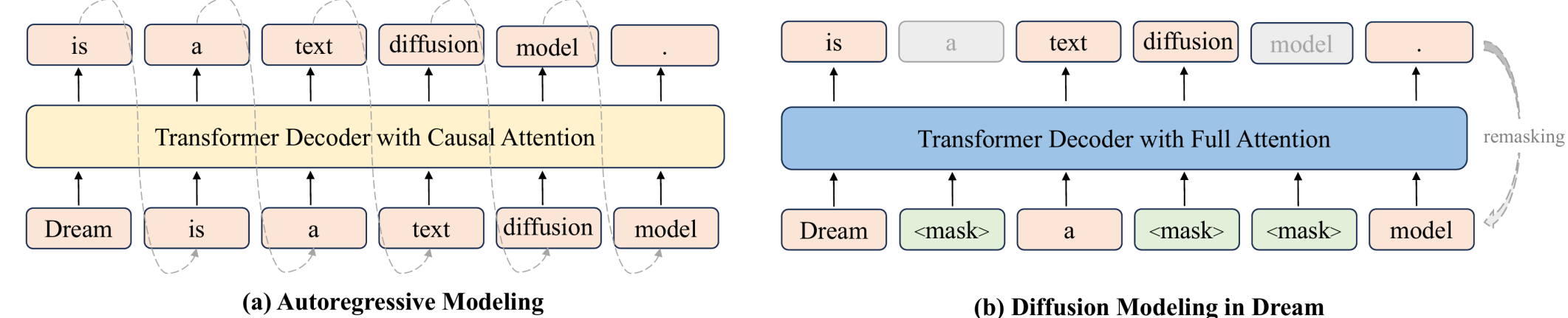


Explore the
Feature Atlas

COLM 2026
NonAR-LM Workshop

1. AR vs. diffusion

- **AR**: sequential next-token decoding
- **DLM**: parallel prediction, iterative unmasking

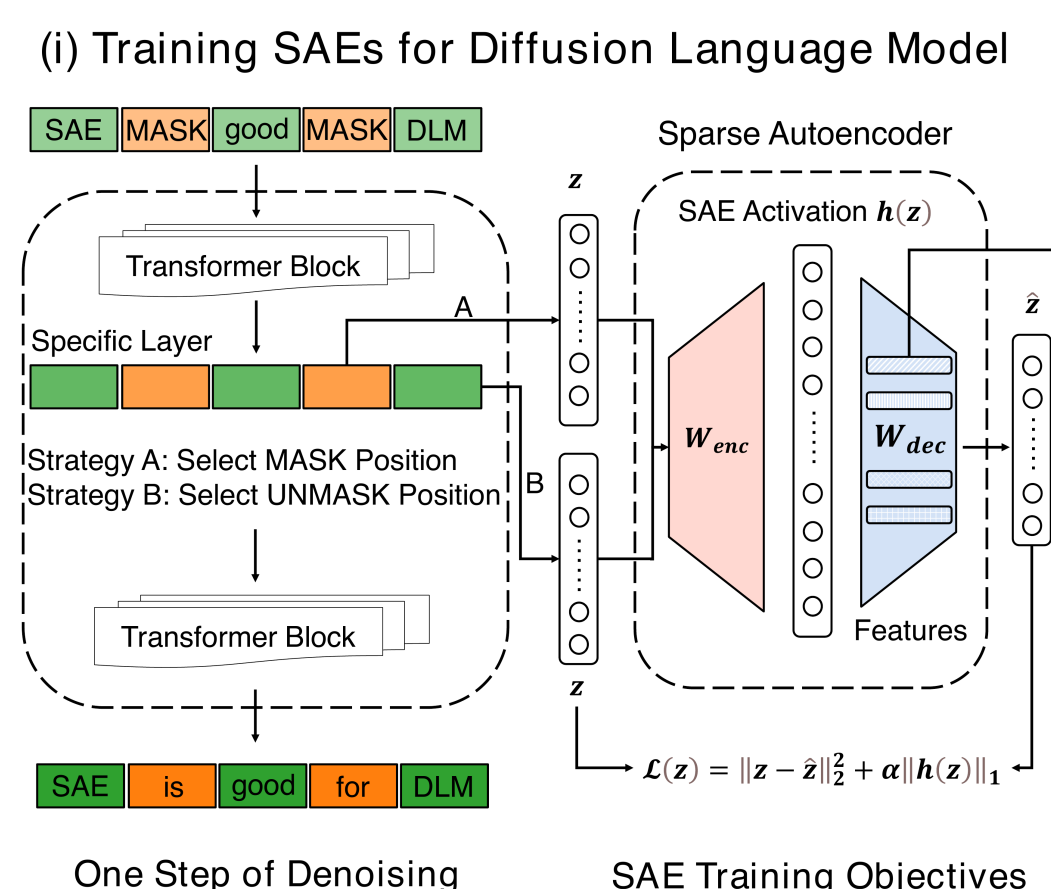


Dream: Ye et al. (2025), Fig. 2.

How do features evolve during denoising?

2. Sparse features with DLM-Scope

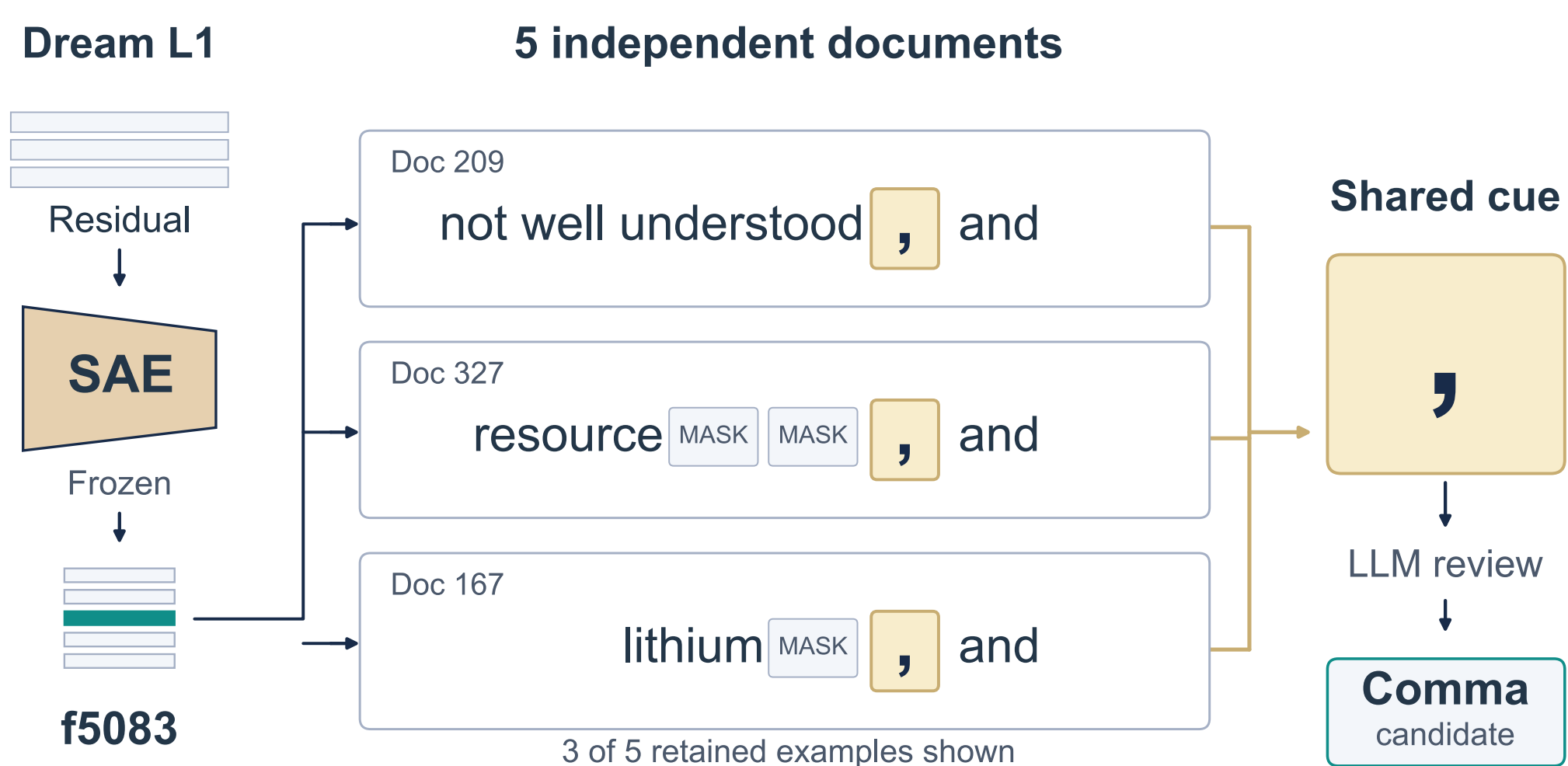
- Frozen **Top-K SAEs**
- **6 layers** x 2 dictionaries
- **196,608 features/model**
- MASK + revealed positions



DLM-Scope: Wang et al. (2026), Fig. 2 (left).

3. Feature autointerpretation

393,216 records | 240 agent-reviewed features



Actual activation inputs → shared cue → candidate label

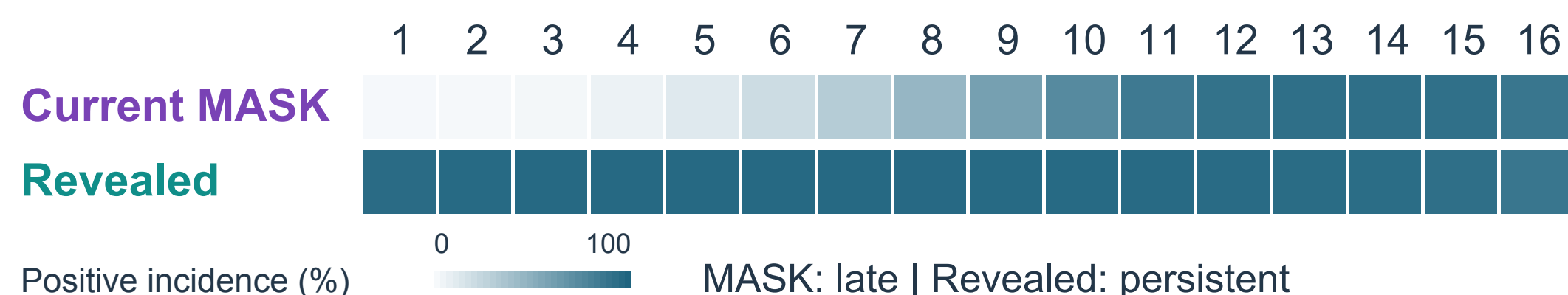
4. Tracking features through time

65 forward inputs → 16 bins

- **Activity**: positive events / actual exposure
- **Comparison**: MASK and revealed equally weighted
- **Mapping**: fixed features, one UMAP per model

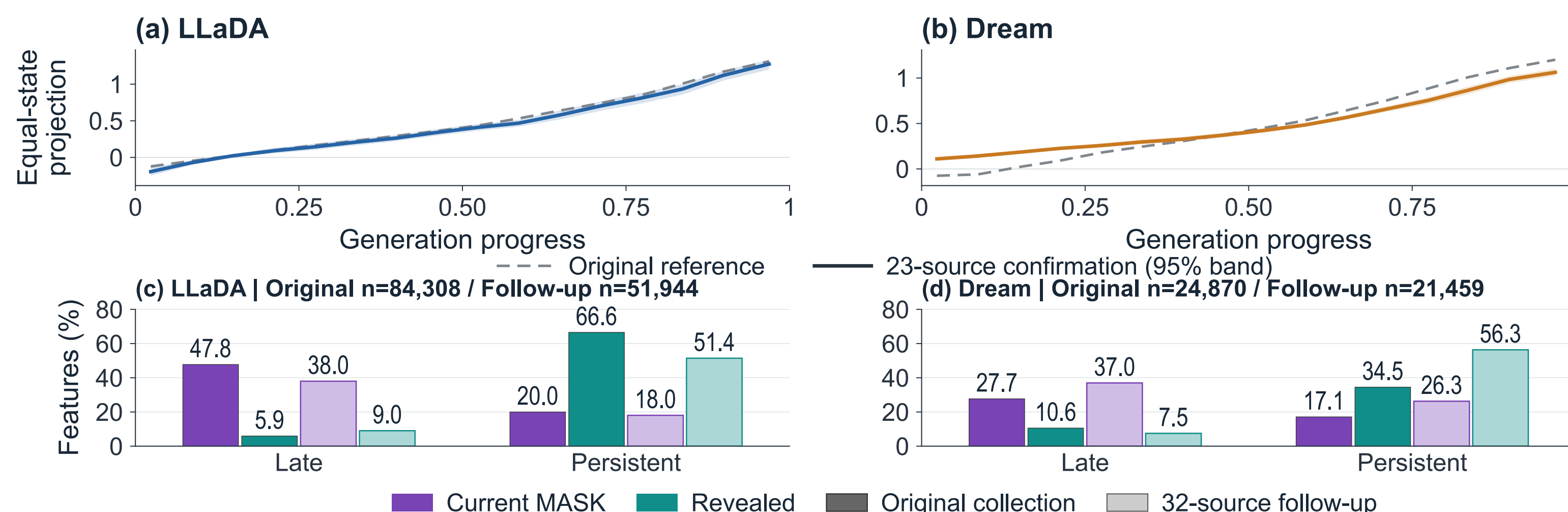
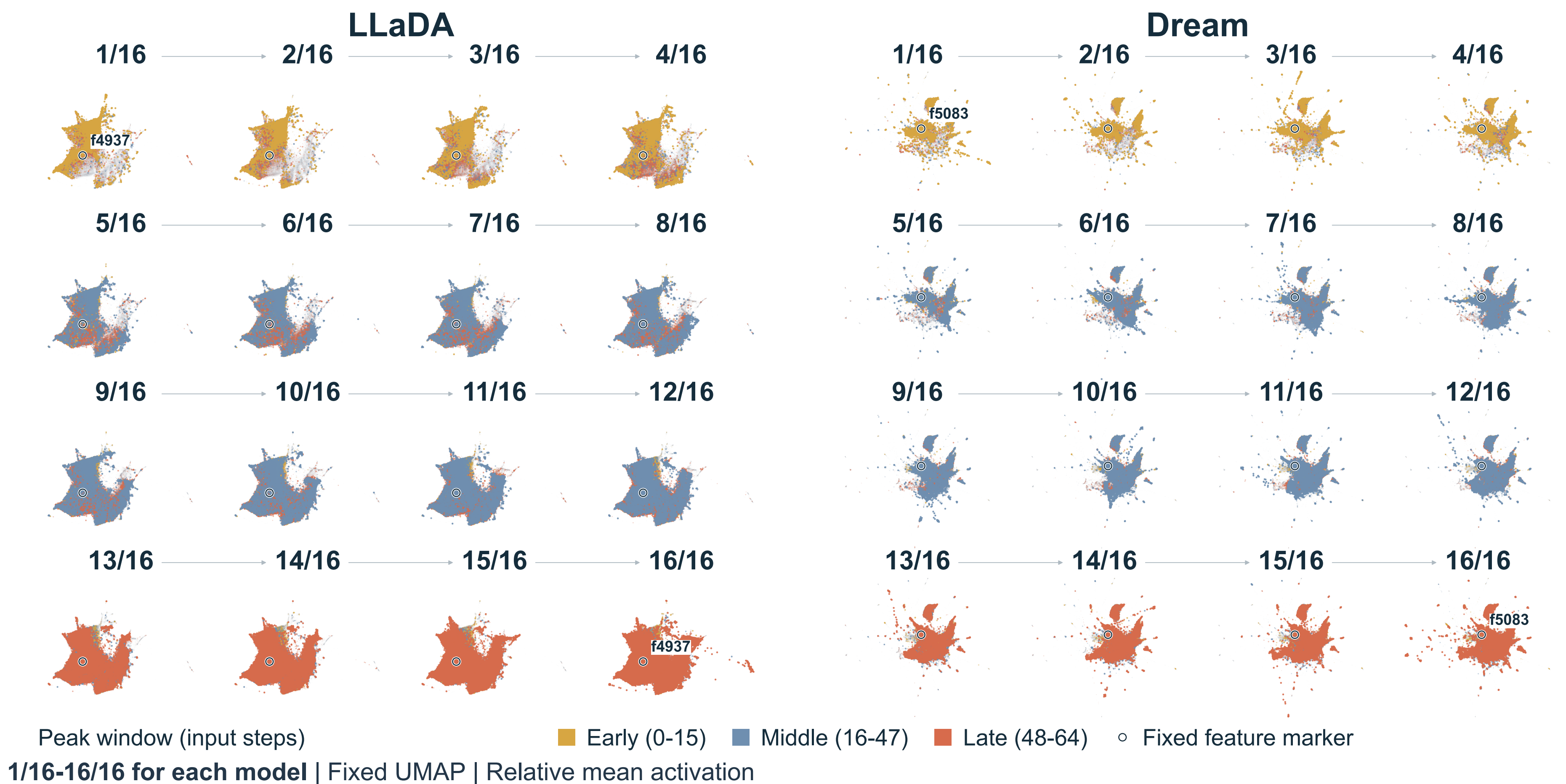
Same feature, different states

LLaDA · L1 MASK-trained SAE · f4937



5. Shared patterns across models

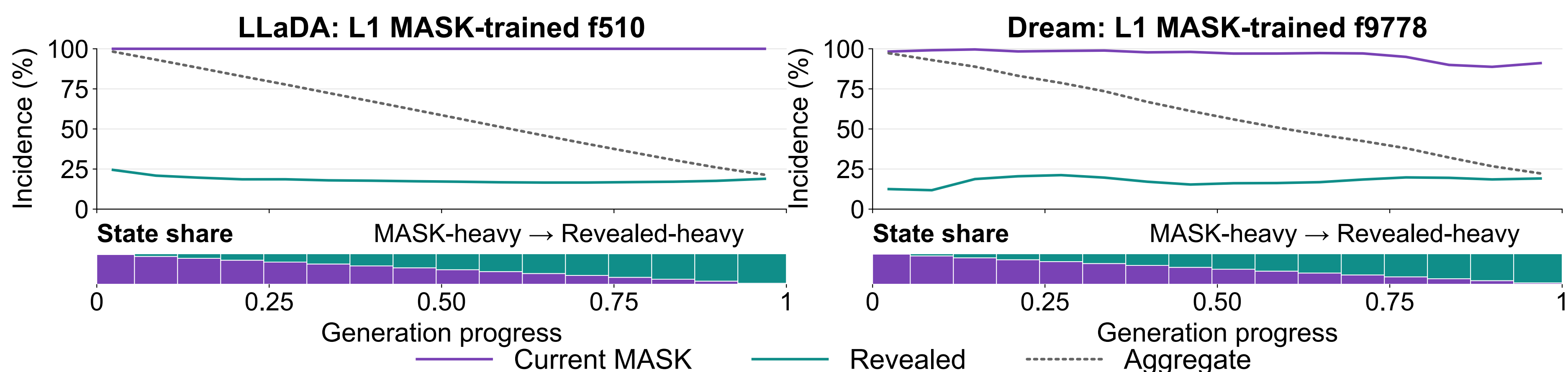
- **MASK**: more late profiles
- **Revealed**: more persistent profiles



Model	Temporal span D reference = 1	95% interval	Activity variance along temporal axis	Positive shifts sources with D > 0
LLaDA	0.959	[0.886, 1.023]	60.8%	23 / 23
Dream	0.689	[0.650, 0.729]	39.6%	23 / 23

6. Persistence and visibility

Persistent conditional responses | Falling aggregate activity



Decline explained by changing MASK share

98.5%

Decline explained by changing MASK share

94.9%

Two feature examples | Share of aggregate decline